

Unit 3.6 Differentiation of logarithmic functions

Note Title

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In this unit, we will find the following derivatives:

$$(1) \frac{d}{dx} \ln x \quad \text{and} \quad \frac{d}{dx} \log_b x = \frac{1}{x \ln b} \quad (\text{p. 218})$$

$$(2) \frac{d}{dx} \ln |x| \quad (\text{p. 220}) \quad \text{See example 6.}$$

Here we use logarithmic differentiation { (3) $\frac{d}{dx} (f(x)^{g(x)})$ (See example 8, p. 222)
(4) Complicated functions.

$$(1) \text{ Find } \frac{d}{dx} (\ln x) \quad y = \ln x \Rightarrow x = e^y$$

Using implicit differentiation, we can show that

$$\frac{d}{dx} (\ln x) = \frac{1}{x}.$$

$$\frac{d}{dx} \ln f(x) = \frac{f'(x)}{f(x)} \quad \text{See examples 2, 3 p. 219}$$

* Logarithmic Differentiation — See steps on p. 221.

Example: (1) Find $\frac{dy}{dx}$ if $y = x^x$ (Form: $y = f(x)^{g(x)}$)

To differentiate a function of the form $y = f(x)^{g(x)}$, we first simplify the function by taking logarithms both sides:

$$\ln y = \ln x^x$$

$$= x \ln x$$

Now we differentiate a product.

$$\Rightarrow \frac{d}{dx} (\ln y) = \frac{d}{dx} (x \ln x)$$

$$\Rightarrow \frac{1}{y} \frac{dy}{dx} = 1 \cdot \ln x + x \times \frac{1}{x} = \ln x + 1.$$

$$\Rightarrow \frac{dy}{dx} = y (\ln x + 1)$$

$$\Rightarrow \frac{dy}{dx} = \underline{x^x (\ln x + 1)}$$

(2) We also use logarithmic differentiation when we differentiate a complicated function, e.g.

$$y = \frac{\sin(x) e^x}{x^3 \sqrt{1+x^2}}$$

Before we differentiate, we first simplify the function, by taking logarithms both sides:

$$\ln y = \ln \left(\frac{\sin(x) e^x}{x^3 \sqrt{1+x^2}} \right)$$

$$= \ln(\sin(x)) + \ln(e^x) - \ln(x^3 \sqrt{1+x^2})$$

$$= \ln(\sin(x)) + x \ln e - (\ln x^3 + \ln(1+x^2)^{\frac{1}{2}})$$

$$= \ln(\sin(x)) + x - (3 \ln x + \frac{1}{2} \ln(1+x^2))$$

$$= \ln(\sin x) + 2 - 3 \ln x - \frac{1}{2} \ln(1+x^2)$$

Now we will differentiate both sides.

$$\frac{d}{dx}(\ln y) = \frac{d}{dx} \left(\ln(\sin x) + 2 - 3 \ln x - \frac{1}{2} \ln(1+x^2) \right)$$

$$\Rightarrow \frac{1}{y} \frac{dy}{dx} = \frac{\cos x}{\sin x} + 1 - \frac{3}{x} - \frac{\frac{1}{2}(0+2x)}{1+x^2}$$

$$= \cot(x) + 1 - \frac{3}{x} - \frac{x}{1+x^2}$$

$$\Rightarrow \frac{dy}{dx} = y \left(\cot(x) + 1 - \frac{3}{x} + \frac{x}{1+x^2} \right)$$

$$= \frac{\sin(x) e^x}{x^3 \sqrt{1+x^2}} \left(\cot(x) + 1 - \frac{3}{x} + \frac{x}{1+x^2} \right).$$



Substitute back!

See examples 7, 8 in textbook, p. 220, 222.

It is important to show all your steps.

See also Video on logarithmic differentiation.

Lastly: Take note of the representation of the number e as a limit. (p. 222)